INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Project Management

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____

RKNEXORA PRIVATE LIMITED
RKNEXORA
PRIVATE LIMITED
GAYA, BIHAR - 805236, INDIA

_____ _____

Signature of Student **Seal & Signature of IPO**

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Project Management successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample project documents and planning templates, and the supervisor's patient and detailed feedback helped me understand how projects are defined, planned, tracked and closed behind the scenes.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Project Management is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

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PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Project Management at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Whether it is launching a new batch in a coaching centre in Gaya, organising a college event, building a website or constructing a bridge, every such piece of work is a project: it has a goal, a deadline, a budget and a team. Many projects fail not because people are lazy but because the work was not planned and tracked well. Project Management is the set of methods that prevents this: defining the project, breaking it into tasks, scheduling it, assigning resources, managing risk and cost, communicating with stakeholders, using agile methods where suitable and tracking progress with tools. This internship offered a first-hand introduction to how these methods work together.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that the report gives a clear and honest account of the internship experience. All examples in it use dummy data and fictitious organisations; no real client, team or financial information was handled and no confidential document was used.

EXECUTIVE SUMMARY

This report documents the internship completed in Project Management at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Project Management means applying knowledge, skills, tools and techniques to a temporary piece of work so that it delivers its objectives within the agreed scope, time, cost and quality. Its great strength is that progress can be measured: how much work is complete, how much money is spent, which risks are open and whether the deadline is safe. A good project manager therefore combines planning and organisation with communication, leadership and numbers. For a student from any discipline, the internship shows how planning, people skills, arithmetic and ethics come together in a real project job.

Objectives. The internship aimed to (i) build a structured understanding of the project lifecycle and of project charter and scope, (ii) learn to create a work breakdown structure, a Gantt chart schedule and a resource plan, (iii) learn risk management and budget and cost control, (iv) learn stakeholder communication, agile and Scrum basics and project tracking tools, (v) complete a project planning and tracking project with charts and diagrams, and (vi) develop communication, documentation and professional discipline.

Work carried out. The student first studied the fundamentals of Project Management, then worked through the project lifecycle, project charter and scope, work breakdown structure, scheduling with Gantt charts, resource planning, risk management, budget and cost control, stakeholder communication, Agile and Scrum basics, and project tracking tools. Each topic had a concept session followed by a hands-on task using free tools, planning templates and dummy data. The later part of the programme was devoted to a case study, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled *Project Planning & Tracking Project*, in which a project charter, a work breakdown structure, a Gantt chart and critical path, a resource and budget plan, a risk register and a tracking board were prepared for a fictitious project and a status report was written.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of how a project is planned, scheduled, controlled and closed, practical familiarity with charters, work breakdown structures, Gantt charts, critical path calculation, resource plans, risk registers, budgets and earned value, communication plans, sprints and Kanban boards, awareness of professional ethics and contract and data protection rules, and improved communication, organisation and professional discipline.

Key observations. Projects succeed when scope is agreed in writing before work starts; the critical path, not the total number of tasks, decides the end date; risks should be listed and owned before they become problems; progress must be measured against a baseline; and honest, regular communication prevents surprises.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Project Management

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand the project lifecycle - initiation, planning, execution, monitoring and closure (Week 1).
2. Learn project charter and scope - charter, objectives, deliverables and scope boundaries (Week 1).
3. Learn the work breakdown structure - breaking a project into tasks and work packages (Week 2).
4. Learn scheduling with a Gantt chart - task durations, dependencies, timeline and critical path (Week 2).
5. Learn resource planning - allocating people, tools and budget to tasks (Week 2).
6. Learn risk management, budget and cost control, and stakeholder communication (Week 3).
7. Learn Agile and Scrum basics and project tracking tools, and complete the project (Week 4).
8. Develop professional discipline - communication, honesty, documentation and report writing (throughout).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the purpose of the topic, its main terms and how it connects with the other parts of Project Management. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - writing a project charter and scope statement, drawing a work breakdown structure, building a Gantt chart in MS Excel, calculating the critical path, preparing a resource and RACI plan, keeping a risk register, calculating budget variance, planning stakeholder communication, running a mock sprint and tracking tasks on a Kanban board. Only fictitious projects, sample documents and dummy figures were used. Only fictitious organisations, sample documents and dummy data were used, so that no real personal or confidential information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, drafts and reports were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Project Management - Concepts Studied

3.1 Project lifecycle

A project is a temporary effort with a defined beginning and end that creates a unique product, service or result. It differs from routine operations, which are repeated and continuing. Every project has a *sponsor* who funds it and gives authority, a *project manager* who is responsible for delivery, a *team* that does the work and *stakeholders* who are affected by it. Success is judged against the *triple constraint* of scope (what is delivered), time (when) and cost (how much), balanced with quality; changing one of them usually affects the others.

The project lifecycle describes the phases through which a project passes: *initiation* (the idea is justified and a charter is approved), *planning* (scope, schedule, budget, resources, risks and communication are planned), *execution* (the work is done and the team is managed), *monitoring and control* (progress is measured against the plan and corrections are made, continuing through all phases) and *closure* (deliverables are handed over, contracts are closed and lessons are recorded). In a predictive or 'waterfall' approach the phases follow one another; in an agile approach short cycles repeat planning, doing and reviewing. Phase-end reviews, called

gates, let the sponsor decide whether to continue.

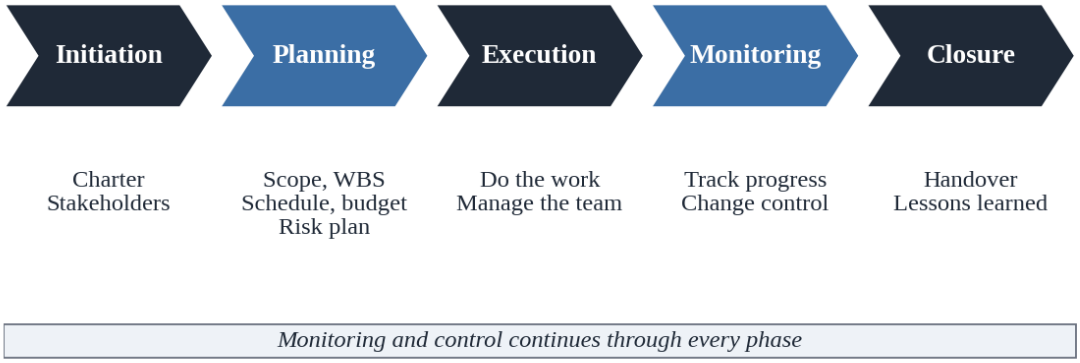


Figure 1: The five phases of the project lifecycle

3.2 Project charter and scope

The *project charter* is a short document that formally authorises the project. It states the purpose and business reason, measurable objectives, main deliverables, high-level scope, assumptions and constraints, major risks, the budget and time limits, the sponsor and project manager and the success criteria. Signing the charter gives the project manager the authority to use resources and protects the team from unclear expectations.

Scope is everything the project will deliver and the work needed to deliver it. A scope statement lists the deliverables, what is included, what is clearly excluded and how acceptance will be judged. Requirements are collected through interviews, surveys, workshops and observation. Without firm boundaries, *scope creep* occurs: small additions accumulate and push up time and cost. It is controlled through *change control*, in which every change request is written, its effect on scope, time, cost and risk is assessed and an authorised person approves or rejects it. The approved scope statement, the work breakdown structure and its dictionary together form the scope baseline.

3.3 Work Breakdown Structure

A Work Breakdown Structure (WBS) breaks the total scope of a project into smaller and smaller parts until it reaches *work packages*, pieces of work small enough to estimate, assign and control. It is organised by deliverables, not by activities, and it follows the *100% rule*: the lowest levels must add up to all of the work and nothing more. A common guide is that a work package should take roughly between a few days and a few weeks, so that progress is visible.

Each element receives a code (1, 1.1, 1.1.1) and a *WBS dictionary* describes it: what is included, who is responsible, the expected duration and cost, and the acceptance criteria. The WBS is the base for the other plans: tasks are listed from it, durations and costs are estimated for each package, resources are assigned and the schedule is built. It can be drawn as a tree or an indented list or built in MS Excel. Common mistakes are listing tasks without deliverables, breaking down too far or too little, and forgetting management work such as meetings and reviews.

3.4 Scheduling (Gantt chart)

A schedule shows when each activity starts and finishes. Building it needs four steps: list the activities from the WBS, estimate the duration of each, find the *dependencies* between them and put them on a calendar. The usual dependency is *finish-to-start* (B cannot start until A finishes); others are start-to-start, finish-to-finish and start-to-finish. A *lead* lets a task start before its predecessor ends and a *lag* adds waiting time. A *milestone* is an event of zero duration, such as 'approval received'. Duration estimates may come from experience, from similar projects or from a three-point estimate (optimistic, most likely and pessimistic), where the expected value is (optimistic + 4 × most likely + pessimistic) / 6.

A *Gantt chart* shows the activities as horizontal bars on a time scale and is the most common schedule display. The *Critical Path Method* finds the longest chain of dependent activities, which decides the shortest possible project duration. A *forward pass* gives the earliest start and finish of each activity, and a *backward pass* gives the latest start and finish. *Float* or *slack* is the time an activity can be delayed without delaying the project; activities on the critical path have zero float, so any delay in them delays the project. If the schedule is too long, it can be shortened by *crashing* (adding resources at a cost) or *fast-tracking* (doing activities in parallel at a risk). The approved schedule becomes the baseline against which progress is measured.

3.5 Resource planning

Resource planning decides who and what is needed for each activity. Resources include people with the right skills, equipment and tools, materials, facilities and money. For each work package the plan states the type and quantity of resources, the period of use, the cost and the availability. A *RACI matrix* makes responsibility clear by marking, for each task, who is Responsible (does the work), Accountable (has final answer), Consulted (gives input) and Informed (is told).

A *resource histogram* shows the load on each resource over time and reveals *over-allocation*, where one person is planned for more work than the available time. It is corrected by *resource levelling* (moving tasks within their float or extending the schedule) or by adding or changing resources. Team members pass through stages - forming, storming, norming, performing and adjourning - and good project managers support each stage with clear goals, fair treatment and recognition. Items to be bought from outside suppliers need a simple procurement plan with quotations, a clear order and an agreed delivery date.

3.6 Risk management

A *risk* is an uncertain event that, if it happens, affects the project objectives positively or negatively; an *issue* is a problem that has already happened. Risk management has five steps: plan the approach; *identify* risks by brainstorming, checklists, SWOT analysis, expert advice and lessons from earlier projects; *analyse* them; plan *responses*; and *monitor* them through the project. Each risk is written in the form 'because of a cause, an event may happen, leading to an effect' and is entered in a *risk register* with an owner.

In *qualitative analysis* each risk is rated for probability and impact, for example on a scale of 1 to 5, and the score is probability × impact. A *probability-impact matrix* shows which risks need attention first. *Quantitative analysis* uses numbers, such as the *expected monetary value* (EMV) = probability × cost of impact. Responses to threats are *avoid* (remove the cause), *mitigate* (reduce probability or impact), *transfer* (for example insurance or a contract) and *accept* (do nothing, or keep a reserve); responses to opportunities are exploit, enhance, share and accept. A *contingency reserve* covers identified risks, and the register is reviewed regularly because risks change.

3.7 Budget and cost control

A project budget starts with cost estimates for each work package, using methods such as analogous estimating (from similar projects), parametric estimating (rate × quantity), bottom-up estimating (adding up the packages) and three-point estimating. Direct costs (such as materials and wages) and indirect costs (such as shared overheads) are included, and a contingency is added for risks. The approved budget spread over time is the *cost baseline*.

Cost control compares what has been spent with what was planned and with the work actually completed. *Earned Value Management* uses three values: *Planned Value* (PV, the budgeted cost of the work scheduled to date), *Earned Value* (EV, the budgeted cost of the work actually completed) and *Actual Cost* (AC). Then *cost variance* CV = EV − AC, *schedule variance* SV = EV − PV, *cost performance index* CPI = EV / AC and *schedule performance index* SPI = EV / PV. An index below 1 means over cost or behind schedule. The *estimate at completion* EAC = BAC / CPI, where BAC is the budget at completion. Control actions include re-planning, removing scope by agreement, using reserves and approving changes only after assessing their cost.

3.8 Stakeholder communication

Stakeholders are the people and groups who affect or are affected by the project: sponsor, customers, team, suppliers, regulators and the community. *Stakeholder analysis* lists them, notes their interest and influence and decides how to engage them. A *power-interest grid* classifies them: high power and high interest are managed closely, high power and low interest are kept satisfied, low power and high interest are kept informed, and low power and low interest are monitored.

A *communication plan* states what information goes to whom, how often, by which channel and who is responsible. Common channels are meetings, written status reports, e-mail, messaging groups and dashboards. Communication becomes harder as teams grow: the number of communication channels in a group of n people is $n(n - 1) / 2$. A good *status report* is short, uses a red-amber-green (RAG) status, shows progress against the baseline, lists risks and issues and states the decisions needed. Conflicts are handled early through listening, facts and agreed escalation paths, and all important decisions are recorded in writing.

3.9 Agile and Scrum basics

Agile is a way of managing work in short cycles, delivering something usable early and changing direction as the customer's needs become clearer. It comes from the Agile Manifesto of 2001, which values individuals and interactions, working products, customer collaboration and responding to change, while still valuing the items on the other side. Agile suits projects where requirements are uncertain, such as software, while a predictive plan suits work with fixed requirements, such as construction. Many projects use a mix.

Scrum is the best known agile framework. A Scrum team has a *Product Owner* (decides what is most valuable and manages the product backlog), a *Scrum Master* (helps the team to follow Scrum and removes obstacles) and *Developers* (do the work). The *Product Backlog* is the ordered list of work, often written as user stories and sized in story points. Work is done in *sprints* of one month or less. The events are *Sprint Planning* (choose and plan the work), the *Daily Scrum* (a 15-minute daily check), the *Sprint Review* (show the result) and the *Sprint Retrospective* (improve the way of working). The sprint produces an *increment*. *Velocity* is the amount of work completed per sprint and helps in planning, and a *burndown chart* shows the remaining work during a sprint.

3.10 Project tracking tools

Tracking tools help the team to see who is doing what and how far the work has reached. Spreadsheets such as MS Excel and Google Sheets are enough for small projects and are good for Gantt charts, budgets and risk registers. Online tools such as Trello, Jira, Asana and Microsoft Planner provide boards, task assignment, due dates, comments, notifications and reports, and many have free plans whose features should be checked on the official site.

A *Kanban board* has columns such as To Do, In Progress and Done, and each task is a card that moves across the board. A *work-in-progress (WIP) limit* stops the team from starting too much at once. Cards carry an owner, due date, labels, a checklist and attachments. Dashboards show task counts, overdue items, burndown charts and workload. Good practice is to name tasks clearly, update the board daily, give every task one owner, review it in the team meeting, keep the plan baseline separate from the live board, and give access only to those who need it.

3.11 Rules, ethics and limitations

Project managers must follow laws and professional ethics. Agreements with clients, vendors and team members are governed by the Indian Contract Act, 1872, so scope, price, deadlines and payment terms should be in writing. Personal data of team members, customers and users may be collected and used only with consent and for a stated purpose under the Digital Personal Data Protection Act, 2023, and the Information Technology Act, 2000 governs electronic records and online conduct. Copyright and licence terms of software, templates and images must be respected. Professional codes, such as the code of ethics of the Project Management Institute, stress responsibility, respect, fairness and honesty.

Good practice includes reporting progress and problems honestly, not hiding delays or overspending, avoiding conflicts of interest in choosing vendors, giving credit to team members, protecting confidential project information, following safety and labour rules, and recording decisions and changes. The tools, features, rates and laws mentioned in this report change often, so they must be verified against official sources, and the figures in the examples are dummy values for learning only.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, introduction to Project Management, Project Lifecycle and Project Charter & Scope.
Week 2	Days 7-12	Break down and schedule: Work Breakdown Structure, Scheduling (Gantt Chart) and Resource Planning, each with a concept session and a hands-on task.
Week 3	Days 13-18	Control risk, cost and communication: Risk Management, Budget & Cost Control and Stakeholder Communication.
Week 4	Days 19-24	Agile and tracking: Agile & Scrum Basics, Project Tracking Tools, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student was introduced to Project Management and studied Project Lifecycle and Project Charter & Scope, with a concept day and a practice day for each.

Week 2: In the second week the study moved to Work Breakdown Structure, Scheduling (Gantt Chart) and Resource Planning, each studied in a concept session and then practised on sample tasks.

Week 3: The third week focused on Risk Management, Budget & Cost Control and Stakeholder Communication.

Week 4: In the fourth week the student studied Agile & Scrum Basics and Project Tracking Tools. A case study exercise and a mock practical exercise were also completed in this week.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Project Lifecycle

Concept: Initiation, planning, execution, monitoring and closure. **Practice carried out:** Mapping the phases and deliverables of a sample project, drawing the lifecycle chart and listing the triple constraint trade-offs.

6.2 Project Charter & Scope

Concept: Charter, objectives, deliverables and scope boundaries. **Practice carried out:** Writing a project charter and scope statement with in-scope and out-of-scope items and recording a sample change request.

6.3 Work Breakdown Structure

Concept: Breaking a project into tasks and work packages. **Practice carried out:** Drawing a WBS tree to work package level, numbering it and writing a WBS dictionary entry for each package.

6.4 Scheduling (Gantt Chart)

Concept: Task durations, dependencies and timeline. **Practice carried out:** Building a Gantt chart in MS Excel, calculating earliest and latest times, float and the critical path and drawing the network diagram.

6.5 Resource Planning

Concept: Allocating people, tools and budget to tasks. **Practice carried out:** Preparing a resource table, a RACI matrix and a budget, and removing a resource over-allocation by levelling.

6.6 Risk Management

Concept: Identifying, ranking and responding to project risks. **Practice carried out:** Preparing a risk register, rating risks on a probability-impact matrix and calculating expected monetary value.

6.7 Budget & Cost Control

Concept: Estimating cost, tracking spending and variance. **Practice carried out:** Preparing a cost baseline and calculating PV, EV, AC, variances, CPI, SPI and EAC from sample data.

6.8 Stakeholder Communication

Concept: Stakeholder list, communication plan and status updates. **Practice carried out:** Preparing a stakeholder list and power-interest grid, a communication plan and a one-page status report.

6.9 Agile & Scrum Basics

Concept: Sprints, backlog, stand-ups and roles. **Practice carried out:** Writing user stories, sizing a backlog in story points, planning a sprint and drawing a burndown chart.

6.10 Project Tracking Tools

Concept: Tracking tasks and progress in tools like Trello or Jira. **Practice carried out:** Setting up a Kanban board with cards, owners and due dates, applying a WIP limit and updating progress daily.

7. Rules, Ethics and Good Practice - Points Studied

Project management operates within a legal and ethical framework. Contracts and agreements must follow the Indian Contract Act, 1872 and be clear on scope, price, time and responsibilities. Team and customer data may be used only with consent and for a stated purpose under the Digital Personal Data Protection Act, 2023. Software, templates, images and documents used in a project must respect copyright and licence terms. Safety, labour and environmental rules apply to the project team and to the work.

Good practice includes honest status reporting, written change control, fair and transparent choice of vendors, protecting confidential information, using strong passwords and two-step verification for project tools, giving access to team members only as needed, recording decisions and lessons learned, and never promising what the team cannot deliver.

8. Process Flows Studied

Process mapping was used to understand how work in Project Management moves from start to finish and who is responsible at each step. The main flows studied were:

- **Project lifecycle:** Idea and business case > Charter approval > Planning > Execution > Monitoring and control > Closure and lessons learned.
- **Scope and WBS:** Requirements > Scope statement > WBS > WBS dictionary > Scope baseline > Change control.
- **Scheduling:** Activity list > Durations > Dependencies > Forward and backward pass > Critical path and float > Schedule baseline.
- **Risk:** Identify > Rate probability and impact > Plan response > Assign owner > Monitor > Review and close.

- **Cost control:** Estimate > Budget and cost baseline > Record actual cost > Measure earned value > Calculate variances > Take corrective action.
- **Agile sprint:** Product backlog > Sprint planning > Daily Scrum > Sprint review > Retrospective > Next sprint.

9. Worked Examples with Sample (Dummy) Data

The examples below use a fictitious project of a local coaching centre, Gyan Jyoti Classes in a town in Bihar: launching a new Class 10 evening batch within 17 working days with a budget of Rs. 60,000. All names, numbers and results are dummy figures for learning.

Example A: Project charter and scope

The charter was written as follows. *Project:* New Class 10 Evening Batch Launch. *Sponsor:* the centre director. *Project manager:* the academic coordinator. *Objective:* to enrol at least 30 students and start the batch on project day 17, within a budget of Rs. 60,000. *Deliverables:* approved syllabus and timetable, a trained teacher, a ready classroom and study material, a promotion campaign, demo classes and enrolments. *In scope:* a Class 10 mathematics and science batch from 5 pm to 7 pm. *Out of scope:* a Class 9 batch, online classes and a new branch. *Assumptions:* a classroom is free in the evening and a teacher can be assigned within a week. *Constraints:* the budget and the start date. *Success criteria:* 30 or more enrolments, cost not above Rs. 60,000 and the batch started on day 17. When a parent asked for a Saturday morning batch as well, it was recorded as a change request and decided by the sponsor, not added to the work.

Example B: Work breakdown structure

The project was broken down into five groups with eight work packages: 1.1 Planning (plan and budget approval, syllabus and timetable); 1.2 Resources (teacher assigned, classroom and study material); 1.3 Marketing (promotion campaign); 1.4 Admissions (demo classes, enrolment and fee collection); and 1.5 Closure (batch start and review). Together the eight packages cover 100% of the scope and no more, and each one has an owner and a deliverable. Figure 2 shows the structure.

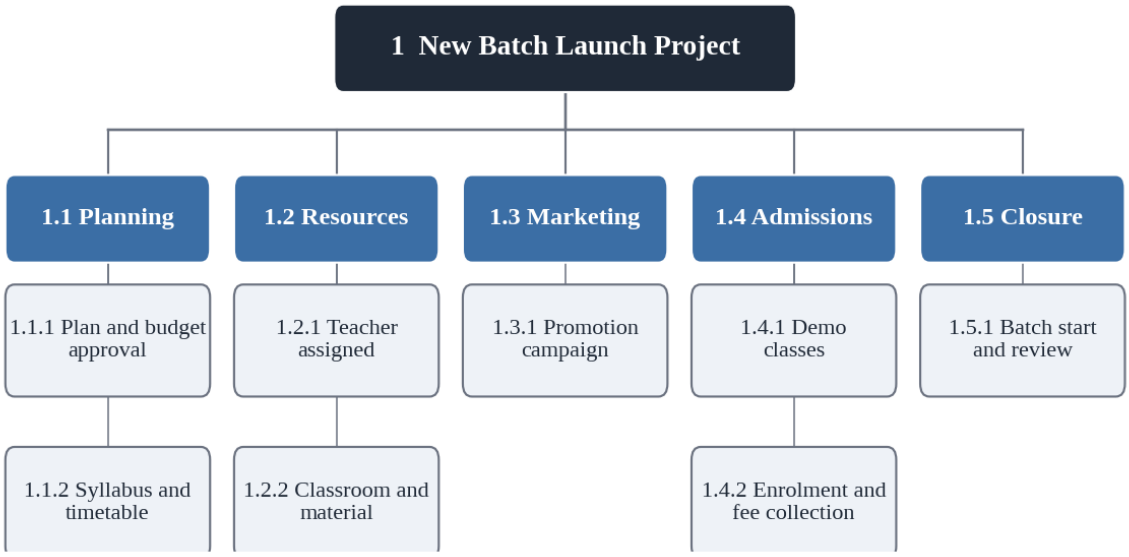


Figure 2: Work breakdown structure of the batch launch project

Example C: Schedule and Gantt chart

The durations and dependencies of the eight packages (named A to H) were listed and the earliest start (ES) and earliest finish (EF) were calculated, with ES of a task equal to the largest EF of its predecessors. Figure 3 shows the Gantt chart; the hatched bars show float.

ID	Task	Days	Predecessor	ES	EF	Float
A	Approve plan and budget	2	-	0	2	0
B	Syllabus and timetable	4	A	2	6	0
C	Teacher assigned	5	A	2	7	2
D	Classroom and material	3	B	6	9	0
E	Promotion campaign	6	A	2	8	1
F	Demo classes	3	C, D, E	9	12	0
G	Admissions and fees	4	F	12	16	0
H	Batch start	1	G	16	17	0

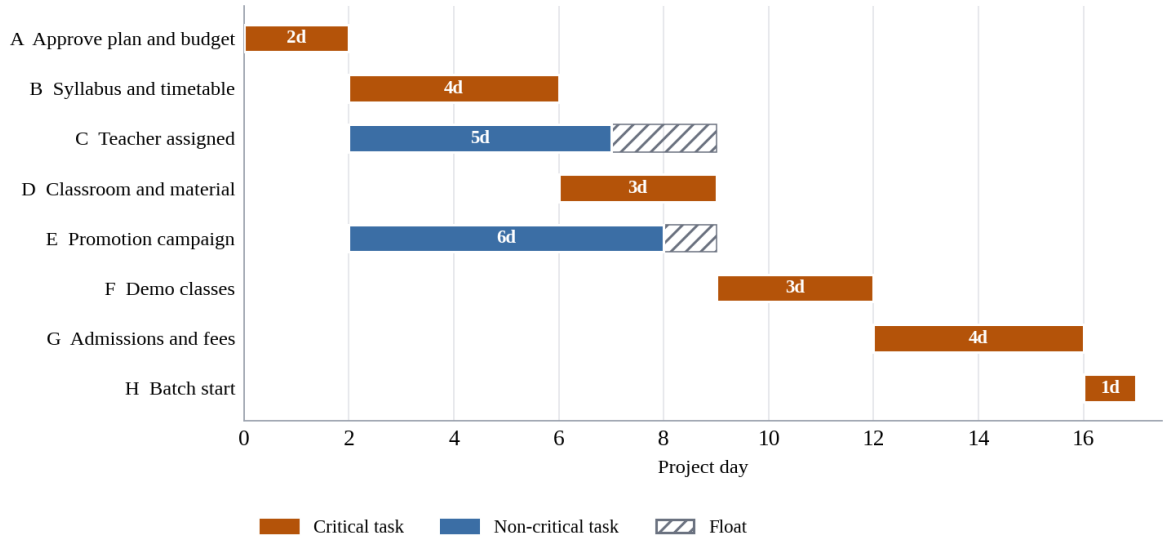


Figure 3: Gantt chart with critical tasks and float

Example D: Critical path

Task F needs C, D and E to finish, and the latest of their finish times is D at day 9 (C finishes at 7 and E at 8), so F starts at day 9. The project finish is day 17. In the backward pass the latest finish of H is 17, G must finish by 16, F by 12 and D by 9. E can finish as late as day 9, so its float = 9 − 8 = 1 day, and C float = 9 − 7 = 2 days. A, B, D, F, G and H have zero float, so the critical path is A, B, D, F, G, H with a length of 2 + 4 + 3 + 3 + 4 + 1 = 17 days (Figure 4). A delay in preparing the classroom (D) would delay the batch start by the same number of days, while a one-day delay in the promotion campaign (E) would not.

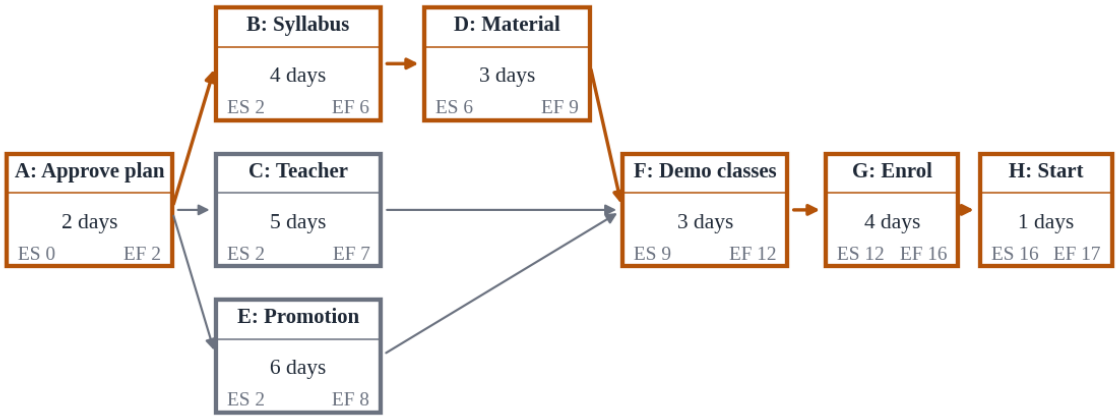


Figure 4: Network diagram with the critical path

Example E: Resource plan

The resources were assigned as shown below. The study material, classroom set-up, promotion and the teacher's first month were budgeted at Rs. 12,000, Rs. 8,000, Rs. 15,000 and Rs. 20,000, which is Rs. 55,000, and a contingency reserve of Rs. 5,000 brought the total to Rs. 60,000.

Task	Resource	Days	Note
A	Director	2	Approves plan and budget
B	Academic coordinator	4	Free after day 6
C	Director and coordinator	5	Interviews and selection
D	Coordinator and printer	3	Study material Rs. 12,000; classroom Rs. 8,000
E	Office staff	6	Promotion Rs. 15,000
F	Teacher and coordinator	3	Teacher cost Rs. 20,000 a month
G	Office staff	4	Enrolment and fee receipts
H	All staff	1	Batch starts

The resource histogram showed that the office staff were planned for task E on days 2 to 8 and also for part of task D on days 6 to 9, an overlap on days 6 to 8. Because the coordinator becomes free when task B ends on day 6, task D was assigned to the coordinator and the overlap was removed without changing the schedule. A RACI matrix recorded the director as accountable for approvals, the coordinator as responsible for the schedule and the office staff as responsible for enrolment.

Example F: Risk register and matrix

Five risks were rated for probability (P) and impact (I) on a scale of 1 to 5. R1: the chosen teacher is not available (P = 2, I = 5, score 10). R2: fewer students enrol than planned (P = 3, I = 4, score 12). R3: printing of study material is delayed (P = 4, I = 2, score 8). R4: spending exceeds the budget (P = 3, I = 3, score 9). R5: power or network failure during demo classes (P = 2, I = 2, score 4). The ranking was R2, R1, R4, R3 and R5 (Figure 5). The expected monetary value of R2, taking a revenue loss of Rs. 15,000 at a probability of 30%, was 0.30 × 15,000 = Rs. 4,500, which is covered by the contingency of Rs. 5,000. Responses were: R2 mitigate by starting promotion early and offering free demo classes; R1 mitigate by keeping a second candidate; R3 avoid by ordering material before day 6; R4 mitigate by weekly cost checks; R5 accept with a printed back-up lesson plan. Each risk had an owner and a review date.

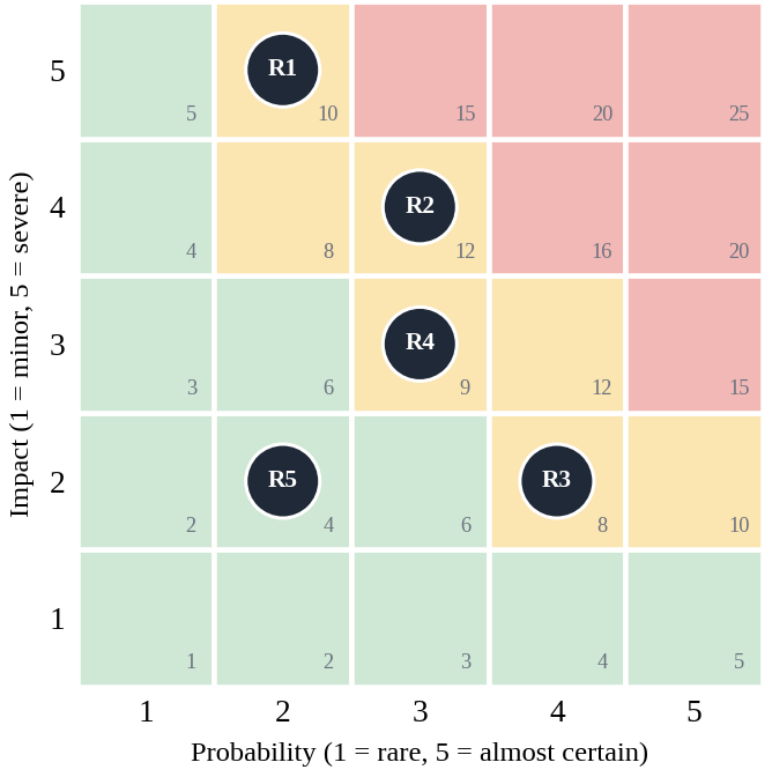


Figure 5: Probability-impact matrix of the five risks

Example G: Budget and earned value

The budget at completion (BAC) of the work packages is Rs. 55,000. At the status date, day 9, the work planned to be complete was worth Rs. 30,000 (PV), the work actually completed was worth Rs. 27,000 (EV) and the money spent was Rs. 31,500 (AC). Then cost variance CV = 27,000 − 31,500 = −Rs. 4,500 (over cost) and schedule variance SV = 27,000 − 30,000 = −Rs. 3,000 (behind schedule). CPI = 27,000 / 31,500 = about 0.86 and SPI = 27,000 / 30,000 = 0.90. If the same cost efficiency continues, the estimate at completion EAC = BAC / CPI = 55,000 / 0.857 = about Rs. 64,167, so the variance at completion VAC = 55,000 − 64,167 = −Rs. 9,167, which is more than the contingency of Rs. 5,000. The team therefore reviewed the promotion costs and reduced printing quantity, and informed the sponsor.

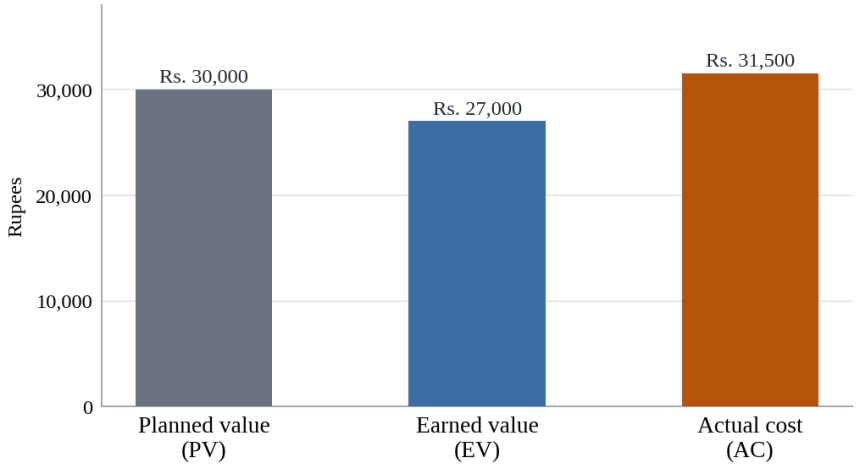


Figure 6: Planned value, earned value and actual cost at day 9

Example H: Stakeholders and communication

The stakeholders were analysed on power and interest (Figure 7). The director has high power and high interest and is managed closely; teachers are managed closely; parents and students have lower power but high interest and are kept informed; local schools have power but low interest and are kept satisfied; and the printer is monitored. A project team of 6 people has 6 × 5 / 2 = 15 communication channels, and adding 2 people raises it to 8 × 7 / 2 = 28, so a short daily meeting and a single messaging group were fixed to avoid confusion. The communication plan is shown below.

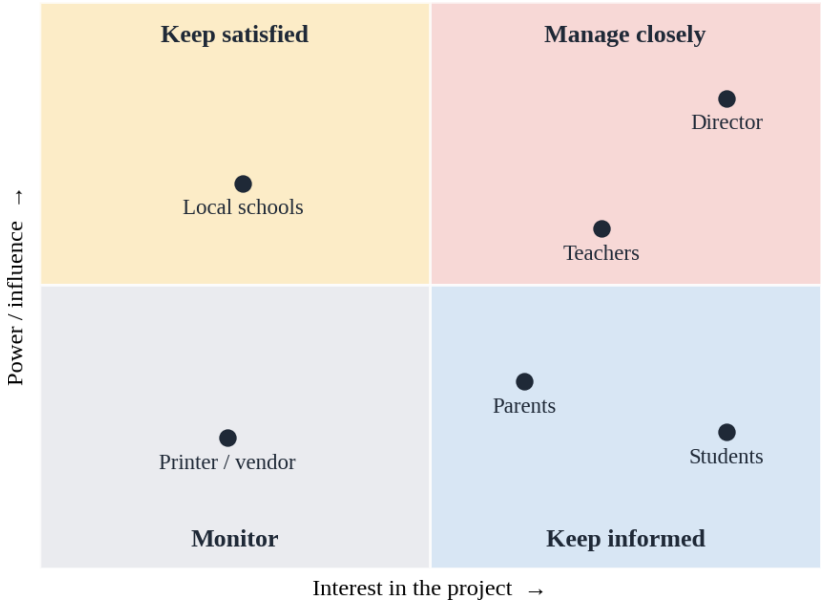


Figure 7: Power-interest grid of the project stakeholders

Stakeholder	Information	Frequency	Channel	Owner
Director	Status report, decisions needed	Weekly	Meeting and report	Project manager
Teachers	Timetable, tasks	Daily	Messaging group	Coordinator
Parents	Demo class dates, fees	As needed	Calls and messages	Office staff
Printer	Order and delivery date	Once	Phone and written order	Office staff

Example I: A sprint and burndown chart

A related ten-day sprint was planned to build the centre's record sheets. The product backlog items were sized in story points: attendance sheet 5, weekly marks sheet 8, parent WhatsApp update template 3, fee tracker 8, progress report format 5, doubt-class register 3, monthly dashboard 5 and back-up and access rules 3, a total of 40 points, which the team committed to. At the sprint review 30 points were complete (attendance sheet, marks sheet, WhatsApp template, fee tracker, doubt-class register and back-up rules), while the monthly dashboard and progress report format (10 points) moved to the next backlog. The velocity was therefore 30 points and the next sprint was planned for about 30 points. The burndown chart (Figure 8) shows that the actual work remaining stayed above the ideal line, and the retrospective noted that two items were started late.

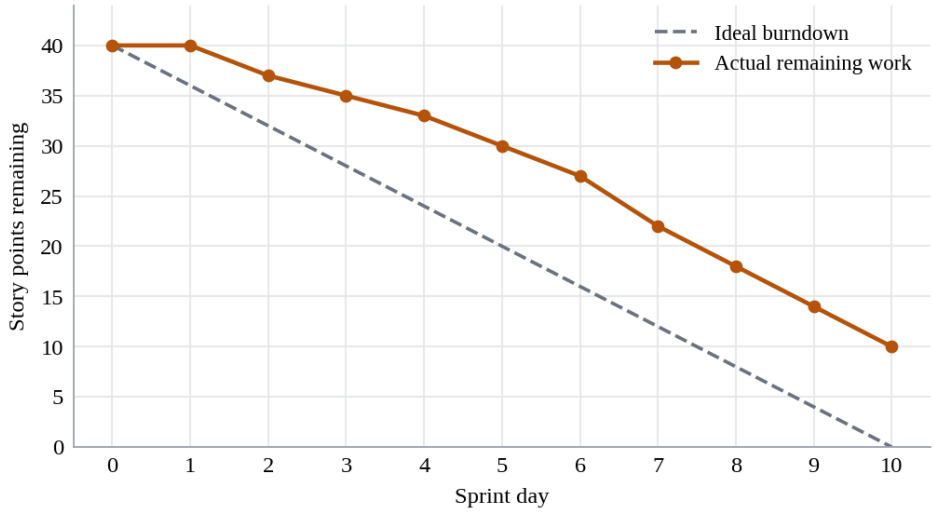


Figure 8: Sprint burndown chart, 40 story points over 10 days

Example J: Tracking board and status report

On a Kanban board of 16 cards, 8 were in Done, 3 in In Progress and 5 in To Do, so 8 / 16 = 50% of the cards were complete, and a work-in-progress limit of 3 was set so that the team finished tasks before starting new ones. The weekly status report used the following summary.

Area	Status	Comment
Scope	Green	No change requests approved; one request pending with the sponsor
Schedule	Amber	SPI 0.90; task D has no float, so close monitoring is needed
Cost	Amber	CPI 0.86; promotion cost reviewed
Risk	Amber	R2 (low enrolment) open; demo class dates fixed to reduce it

The report ended with two decisions needed from the sponsor: approval of the revised printing quantity and a decision on the Saturday batch request.

10. Project: Project Planning & Tracking Project

Scope. The internship work was organised around a project in which a project charter, a work breakdown structure, a Gantt chart with the critical path, a resource and budget plan, a risk register and a tracking board were prepared and a status report was written for a fictitious project, using sample documents and dummy data.

Steps followed. (i) Chose a fictitious project and wrote the charter and scope statement; (ii) prepared the work breakdown structure and its dictionary; (iii) listed activities, durations and dependencies and drew the Gantt chart in MS Excel; (iv) calculated the critical path and float and drew the network diagram; (v) prepared the resource plan and RACI matrix; (vi) prepared the budget, risk register and probability-impact matrix; (vii) calculated earned value measures and prepared a communication plan; (viii) set up a Kanban board and planned a sprint with a burndown chart; and (ix) wrote the status report and reviewed the work with the supervisor.

Skills and tools. WBS, Gantt Chart, Risk Management, MS Excel, Trello or Jira basics and Project Documentation.

Outputs. Project charter and scope statement, WBS and dictionary, Gantt chart, network diagram and critical path, resource plan and RACI matrix, budget and earned value sheet, risk register and matrix, communication plan, sprint plan and burndown chart, Kanban board, status report, and this internship report.

11. Case Study Exercise: A Project That Finished Late and Over Budget

A case study exercise was completed on a typical business problem. The business and figures are illustrative and not taken from any real organisation.

Background. A small training institute (fictitious) decided to launch an online learning portal in 8 weeks with a budget of Rs. 3,00,000. The owner appointed a freelance developer and handled the project himself without a written plan.

Problem observed. The portal was delivered after 15 weeks, a delay of 7 weeks or 87.5%, at a cost of Rs. 4,50,000, an overrun of Rs. 1,50,000 or 50%, and some features still did not work. There was no charter, so the goals were never written down. The owner kept adding features - 14 additional requests in all - without checking their effect on time and cost. There was no work breakdown structure, only one final deadline, so delays were noticed late. No risks were listed, and when the developer fell ill for three weeks nobody could continue the work. Spending was not tracked against a budget, and the owner and developer used different messaging chats, so instructions were missed. Testing was left to the last week. The owner felt that projects cannot be controlled.

Analysis. The main problem was not the developer but the absence of basic project management. Without a charter and scope baseline, change was uncontrolled. Without a WBS and schedule, progress could not be measured. Without a risk register, a predictable risk became a crisis. Without a cost baseline, the overrun was seen only at the end, and poor communication multiplied the confusion.

Recommendations. (1) Write a charter with objectives, scope, budget and success criteria and have it approved before any work. (2) Create a WBS and a schedule with milestones and a critical path. (3) Use a change control process in which every request is assessed for time and cost. (4) Keep a risk register with owners, and arrange a back-up developer. (5) Set a cost baseline and track earned value weekly. (6) Fix a communication plan with a weekly status report and one shared channel. (7) Plan testing in each phase and use short sprints to show working features early.

Expected result and lesson learned. With a clear scope, a tracked plan, managed risk and regular communication, the project is expected to finish close to its plan and any problem to be found early. The lesson is that project management is a system: scope, time, cost, risk and communication must be managed together, and progress must be measured against a baseline.

12. Mock Practical Exercise: A Thirty-Day Mini Project Plan

In the mock practical exercise the student planned and prepared a thirty-day mini project for a sample (dummy) local organisation, from the first charter to the final status report, using the tools and methods practised earlier. The steps followed were:

1. Initiation: wrote the charter, objectives, success criteria and stakeholder list.
2. Scope: wrote the scope statement and set up change control.
3. WBS: broke the scope into work packages and numbered them.
4. Schedule: listed durations and dependencies, drew the Gantt chart and found the critical path.
5. Resources: prepared the resource plan, RACI matrix and budget.
6. Risk: prepared the risk register and probability-impact matrix with responses and owners.
7. Communication and tracking: prepared a communication plan and set up a Kanban board with a sprint plan.
8. Reporting: used sample data to calculate earned value measures and wrote a one-page status report with decisions needed.

The exercise showed that every step depends on the one before it: an unclear scope breaks the WBS, a wrong dependency changes the critical path, unmanaged risks use up the budget, missing tracking hides delay, and an honest status report is the base of the next decision.

13. Sample Checklists Prepared

Checklists were prepared as part of the hands-on tasks. They reduce errors and make routine work easier to review.

Initiation and Scope Checklist

- Business reason and objectives written and measurable.
- Charter approved by the sponsor.
- Stakeholders identified and analysed.
- Scope statement lists inclusions, exclusions and acceptance criteria.
- Change control process agreed.

WBS and Schedule Checklist

- WBS follows the 100% rule and is organised by deliverables.
- Every work package has an owner and a code.
- Durations estimated with a stated method.
- Dependencies and milestones listed; critical path calculated.
- Schedule baseline saved.

Resource and Risk Checklist

- Resources assigned and availability checked.
- RACI matrix completed; over-allocation removed.
- Risks identified with cause, event and effect.
- Probability, impact and score recorded; owner and response assigned.
- Risk register reviewed on fixed dates.

Budget and Cost Control Checklist

- Cost estimates prepared for each work package.
- Contingency stated separately.
- Cost baseline approved.
- Actual cost recorded weekly; PV, EV and AC calculated.
- Variances explained and corrective action recorded.

Communication, Agile and Closure Checklist

- Communication plan states who, what, how often and how.
- Status report uses red-amber-green status and states decisions needed.
- Product backlog ordered; sprint goal and velocity recorded.
- Board updated daily; each task has one owner.
- Deliverables handed over, accounts closed and lessons learned recorded.

14. Tools and Resources

Learning materials, MS Excel and Google Sheets for WBS, Gantt charts, budgets and risk registers, free versions of Trello or similar boards for tracking, Canva or draw.io style diagram tools for charts, MS Word for documents, PowerPoint for presentation, sample project templates, official guides of project management bodies for reference, assignments and mentor feedback were used during the internship.

15. Programme at a Glance

The programme was in the domain of Project Management with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

LEARNING OUTCOME

a. Knowledge Acquired

1. The meaning of a project, the triple constraint and the phases of the project lifecycle.
2. How to write a project charter and a scope statement and control scope changes.
3. How to build a work breakdown structure and a WBS dictionary.
4. How to build a Gantt chart, find the critical path and calculate float.
5. How to plan resources, prepare a RACI matrix and level resources.
6. How to identify, rate and respond to risks, and how to calculate budget variance, CPI, SPI and EAC.
7. How to plan stakeholder communication, run a sprint and track work on a board with a status report.

Glossary of Terms Learned

- **Project** - Temporary effort that creates a unique product, service or result.
- **Triple constraint** - Scope, time and cost, balanced with quality.
- **Charter** - Document that formally authorises a project.
- **Scope creep** - Uncontrolled growth of project scope.
- **Change control** - Written process for assessing and approving changes.
- **WBS** - Work Breakdown Structure: breakdown of scope into work packages.
- **Milestone** - Event of zero duration marking progress.
- **Gantt chart** - Bar chart showing tasks against time.
- **Critical path** - Longest chain of dependent tasks that decides the project duration.
- **Float / Slack** - Delay a task can have without delaying the project.
- **Crashing / Fast-tracking** - Shortening the schedule by adding resources / by overlapping tasks.
- **RACI** - Responsible, Accountable, Consulted, Informed.
- **Resource levelling** - Adjusting the schedule to remove over-allocation.
- **Risk register** - List of risks with ratings, owners and responses.
- **EMV** - Expected monetary value: probability multiplied by impact cost.
- **PV / EV / AC** - Planned value / earned value / actual cost.
- **CPI / SPI** - Cost / schedule performance index: EV divided by AC / by PV.
- **BAC / EAC** - Budget at completion / estimate at completion.
- **Sprint / Backlog** - Short work cycle in Scrum / ordered list of work to be done.
- **Kanban / WIP** - Board of task cards by status / limit on work in progress.

b. Skills Learned

Technical skills. Writing project charters and scope statements; building a WBS and dictionary; drawing Gantt charts in MS Excel; calculating critical path and float; preparing resource plans and RACI matrices; maintaining risk registers and probability-impact matrices; preparing budgets and earned value calculations; planning stakeholder communication; planning sprints and drawing burndown charts; using Kanban boards; and writing status reports.

Professional skills. Planning and organising; analytical and numerical thinking; leadership and teamwork; clear communication and report writing; negotiation and conflict handling; attention to detail; time management; ethical and legal awareness; and acceptance of feedback.

These skills were developed through repeated practice and daily documentation. Scheduling and cost exercises improved numerical thinking, risk and stakeholder work improved foresight and empathy, charts

and diagrams improved the ability to explain a plan visually, and writing status reports improved communication.

c. Observed Attitude and Gained Values

1. Plan before acting: a clear plan saves time, money and stress.
2. Accountability: every task and risk needs one clear owner.
3. Transparency and honesty: delays, risks and costs must be reported truthfully.
4. Collaboration: projects succeed through teamwork and respect for stakeholders.
5. Discipline and punctuality: regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Understanding terms such as critical path, float, earned value and story points was handled by using a glossary, worked examples in Excel and mentor guidance.

Finding the correct dependencies and critical path was difficult at first; this was handled by drawing the network diagram by hand and checking the forward and backward pass with the supervisor.

Choosing realistic durations and costs was handled by using ranges, three-point estimates and clear assumptions written beside each estimate.

Learning several tools in a short time was handled by practising each tool on a small task the same day and keeping short step notes for revision.

Maintaining consistent documentation was handled by using daily log entries and reviewing feedback to improve documentation progressively.

Managing time between practice, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Project Coordinator, Junior Project Manager, Project Management Office (PMO) Analyst, Business Analyst, Operations Coordinator, Event or Programme Coordinator, Product Owner trainee and, with further study, Scrum Master and Project Manager. Opportunities exist in IT and software companies, construction and infrastructure, banks and finance, education, healthcare, manufacturing, consulting firms, start-ups and non-profit organisations, and in government projects. Related study options include BBA, MBA or PGDM with project management, postgraduate diplomas in project management, and certifications such as CAPM and PMP from the Project Management Institute, PRINCE2, Scrum certifications and the Google Project Management certificate. Because methods and tools change quickly, continuous learning and practical experience are essential.

CONCLUSION

The internship in Project Management was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how a project is planned, controlled and closed - through the project lifecycle, charter and scope, WBS, scheduling, resource planning, risk management, budget and cost control, stakeholder communication, agile methods and tracking tools - and acquired practical familiarity with charters, WBS diagrams, Gantt charts, critical path calculations, risk registers, earned value measures, sprint plans and status reports. The experience has strengthened the student's employability and provides a foundation for further study and work in management and operations.

Reflection

Looking back, the most useful lesson of the internship was that a project is a chain, and a chain is only as strong as its weakest link. A clear charter sets direction, the WBS shows all the work, the schedule shows the order, resources make it possible, risk management protects it, cost control keeps it affordable, communication keeps people aligned and tracking shows the truth. The case study showed how a project can slip when even one link is missing. Drawing the Gantt chart, the network diagram and the risk matrix made abstract ideas concrete, and supervisor feedback showed where understanding was weak. The habit of recording each day's work also made it easier to see progress over the weeks.

Objectives and Achievement

The objective of understanding the project lifecycle, charter and scope was achieved through concept sessions, a charter and a scope statement. The objective of learning the WBS, scheduling and resource planning was achieved through a WBS diagram, a Gantt chart with critical path and a resource and RACI plan. The objective of learning risk, cost and communication was achieved through a risk register, an earned value calculation and a communication plan. The objective of learning agile methods and tracking tools was achieved through a sprint plan, a burndown chart and a Kanban board. The objective of completing the project was achieved through the status report and this report. The objective of developing professional discipline was achieved through daily attendance, supervisor review and report writing.

Future Scope

The knowledge gained can support further study of management, commerce and technology, and careers in project coordination and management in IT, construction, education, healthcare and public services. With growing infrastructure and digital projects, start-ups, remote and hybrid teams, wider use of agile methods and project tools and greater use of data and artificial intelligence in planning, trained and ethical project professionals are increasingly needed, including in Bihar.

Suggestions for Improvement of the Programme

1. Include live practice with a project tool on a real small project and talks by project managers and local project owners, with appropriate permissions.
2. Continue regular mentor feedback and progress reviews.
3. Provide more project-based activities such as planning and tracking a small real project for a local organisation under supervision.
4. Encourage students to maintain detailed daily documentation.

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ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]